

VPN inštalácia

Špecializované IKT systémy Windows

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Requirements

DC

Nainštalovanú rolu ADDS, DNS, DHCP a post-OS-Installation konfiguráciu (hostname, IP, firewall)

SRV1

Dokončenú post-OS-Installation konfiguráciu (hostname, IP, firewall, domain join)

Client

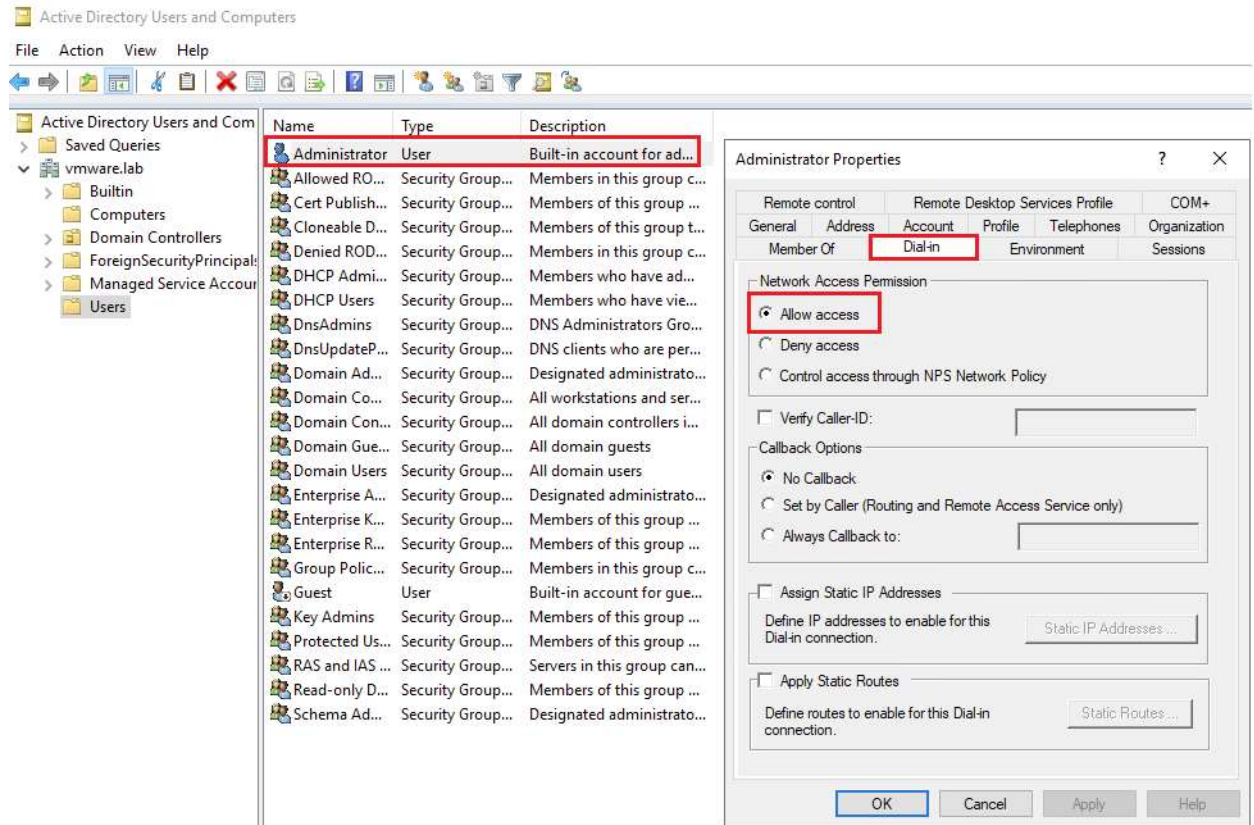
Dokončenú post-OS-Installation konfiguráciu (hostname, IP, firewall, domain join)

Account

<domain>\Administrator

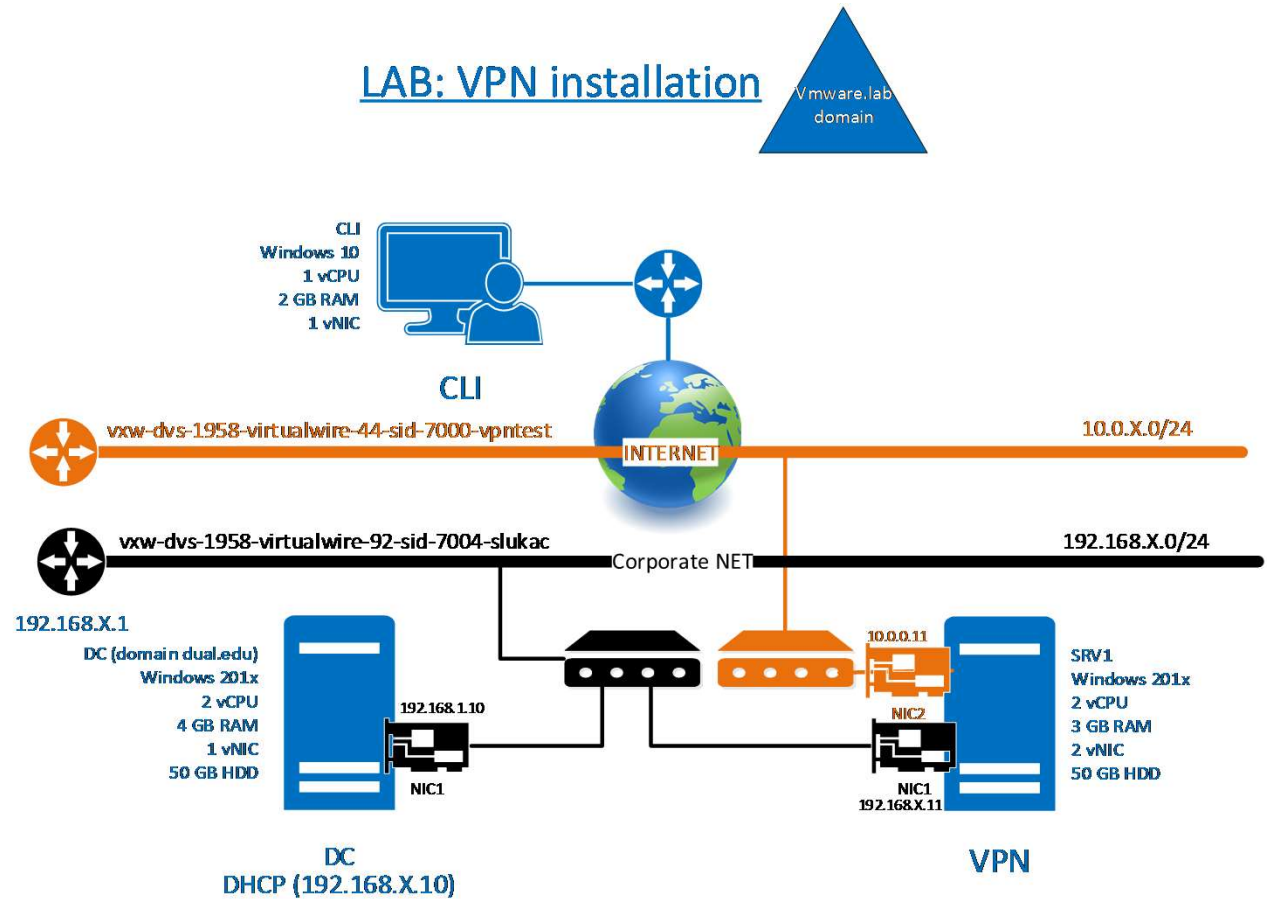
Requirements

<domain>\Administrator musí mať povolený Network Access Permission



Krok 1

::Inštalácia 3 VMs podľa schémy



Krok 2

::Add 2nd vNICs na SRV1 (VPN serveri)

Edit Settings | srv1

Virtual Hardware | VM Options

ADD NEW DEVICE

- > CPU
- > Memory
- > Hard disk 1
- > SCSI controller 0
- > Network adapter 1
- > **New Network ***
- > CD/DVD drive 1

Edit Settings | srv1

Virtual Hardware | VM Options

ADD NEW DEVICE

Select Network

Filter	
Name	Distributed Switch
vm_vlan75	Dual
vMotion	Dual
vxxw-dvs-1958-virtualwire-100-sid-7012-pbarczl3	Mgmt.
vxxw-dvs-1958-virtualwire-101-sid-7013-ppastor	Mgmt.
vxxw-dvs-1958-virtualwire-42-sid-7014-mskara	Mgmt.
vxxw-dvs-1958-virtualwire-44-sid-7000-vpntest	Mgmt.
vxxw-dvs-1958-virtualwire-45-sid-7001-Backup	Mgmt.
34 items	

CANCEL

OK

Krok 2

::Premenovanie vNICs na SRV1 (VPN serveri)



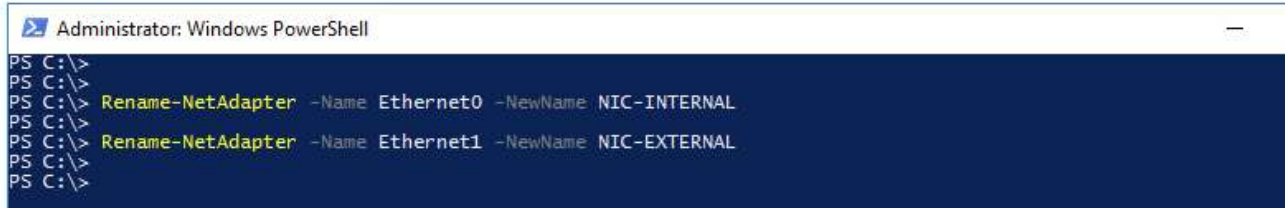
Network Connections

Control Panel > Network and Internet > Network Connections >

Organize ▾ Disable this network device Diagnose this connection Rename this connection View status of this connection Change s

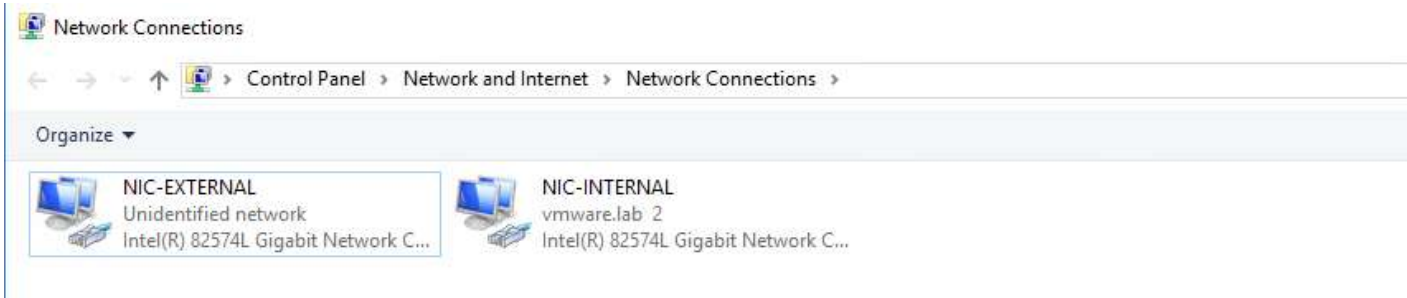
Ethernet0
vmware.lab 2
Intel(R) 82574L Gigabit Network C...

Ethernet1
Unidentified network
Intel(R) 82574L Gigabit Network C...



Administrator: Windows PowerShell

```
PS C:\>  
PS C:\>  
PS C:\> Rename-NetAdapter -Name Ethernet0 -NewName NIC-INTERNAL  
PS C:\> Rename-NetAdapter -Name Ethernet1 -NewName NIC-EXTERNAL  
PS C:\>  
PS C:\>
```

Network Connections

Control Panel > Network and Internet > Network Connections >

Organize ▾

NIC-EXTERNAL
Unidentified network
Intel(R) 82574L Gigabit Network C...

NIC-INTERNAL
vmware.lab 2
Intel(R) 82574L Gigabit Network C...

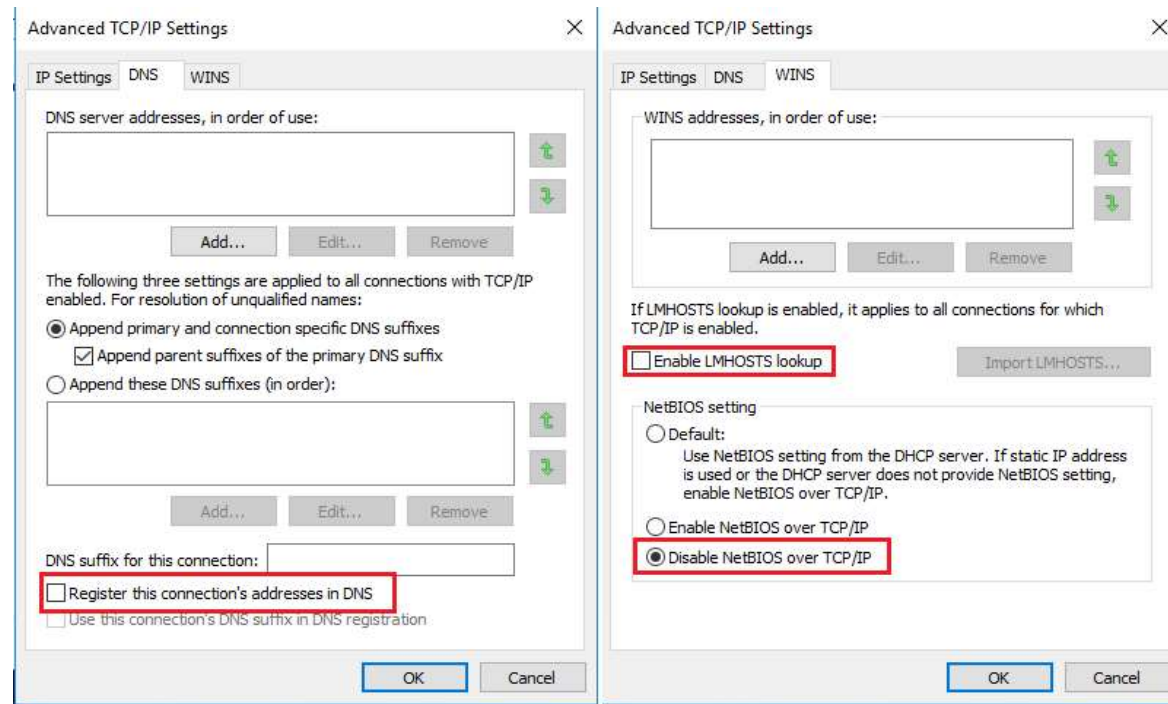
Krok 3

::Overenie sieťovej konfigurácie na SRV1 (VPN server)

	Network	GW	DNS	IPv6
NIC_Internal	192.168.X.0/24	none	10.0.0.10	Enabled
NIC_External	10.0.X.0/24 VLAN 5	10.0.X.1	none	Enabled

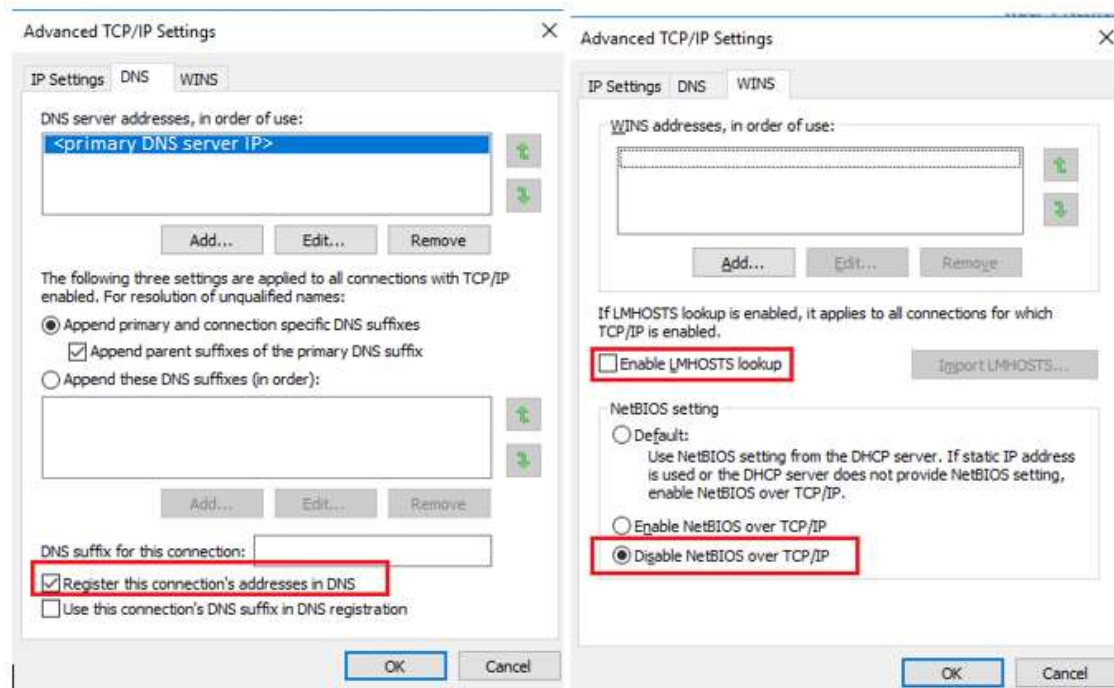
Krok 4

::Vypnutie registrácie DNS a NetBios na Externom NIC SRV1



Krok 5

::Zapnutie registrácie DNS a NetBios na INTERNOM NIC SRV1



Krok 6

::Zapnutie FW na všetkých FW profiloch SRV1

```
Administrator: Windows PowerShell
Windows PowerShell
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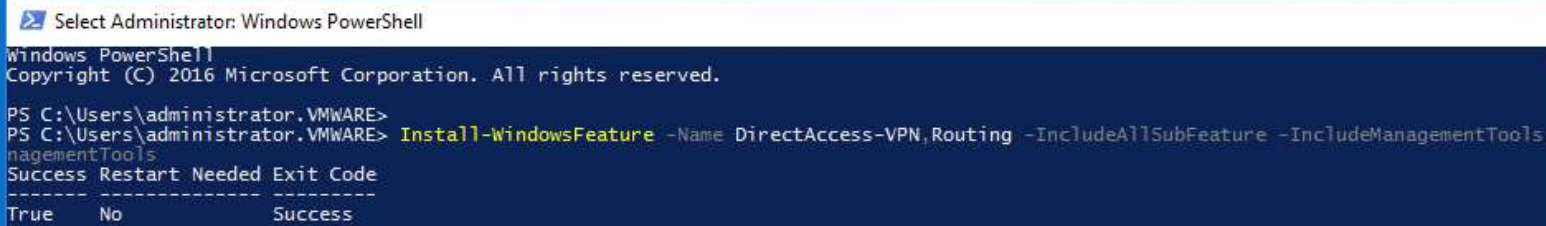
PS C:\Users\administrator.TESTLAB> Get-NetFirewallProfile | ft -AutoSize
```

Name	Enabled	DefaultInboundAction	DefaultOutboundAction	AllowInboundRules	AllowLocalFirewallRules	AllowLocalIPsecRules
Domain	True	Block	Allow	NotConfigured	NotConfigured	NotConfigured
Private	True	Block	Allow	NotConfigured	NotConfigured	NotConfigured
Public	True	Block	Allow	NotConfigured	NotConfigured	NotConfigured

```
PS C:\Users\administrator.TESTLAB>
```

Krok 7

::Inštalácia role



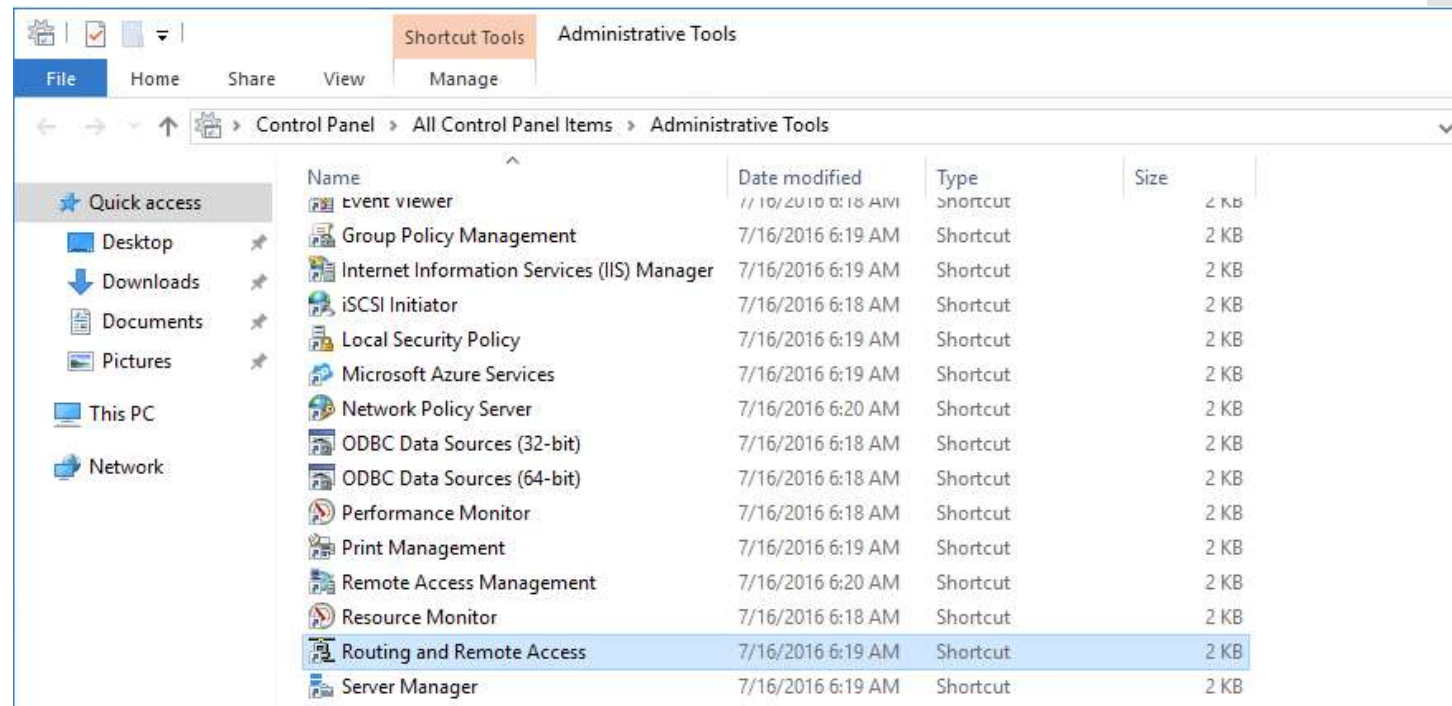
```
Select Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) 2016 Microsoft Corporation. All rights reserved.

PS C:\Users\administrator.VMWARE>
PS C:\Users\administrator.VMWARE> Install-WindowsFeature -Name DirectAccess-VPN, Routing -IncludeAllSubFeature -IncludeManagementTools
Success Restart Needed Exit Code
-----
True      No          Success
```

Krok 8

::Konfigurácia role VPN

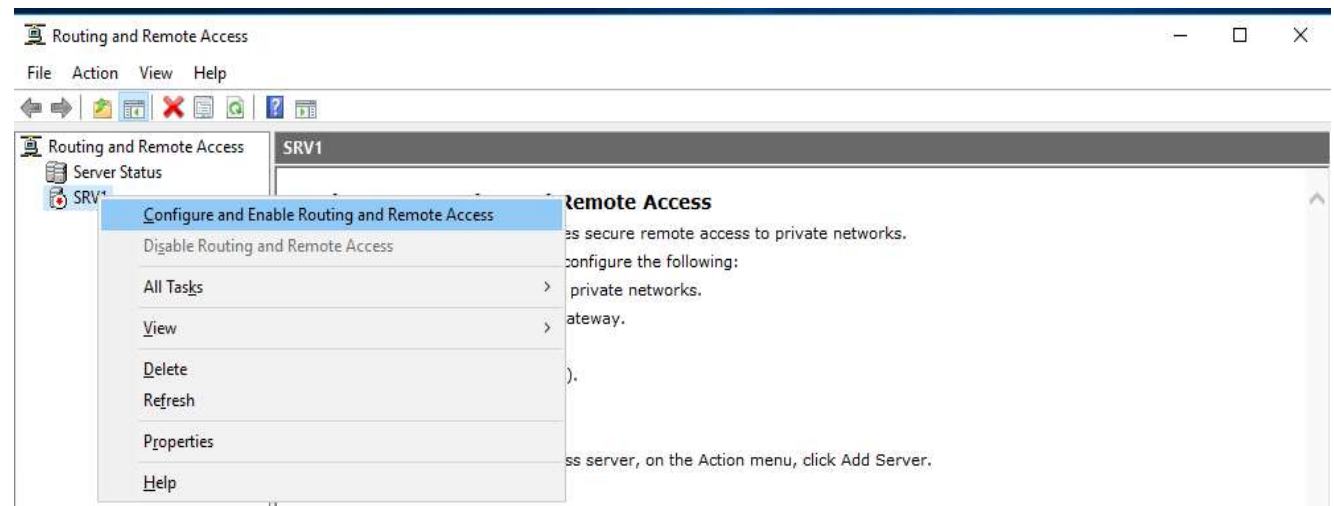
Administrative tools -> Remote Access Management console



Krok 8

::Konfigurácia role VPN

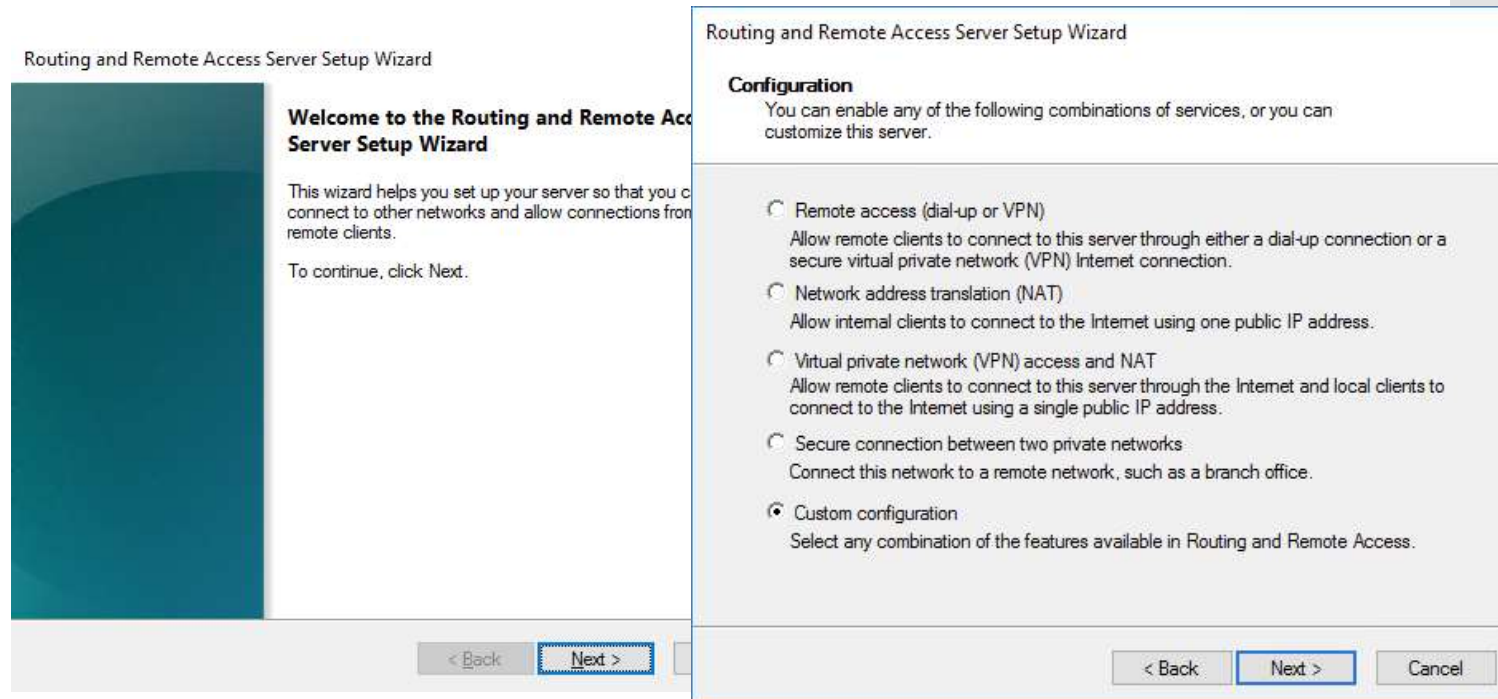
Administrative tools -> Remote Access Management console



Krok 8

::Konfigurácia role VPN

Administrative tools -> Remote Access Management console



Krok 8

::Konfigurácia role VPN

Administrative tools -> Remote Access Management console

Routing and Remote Access Server Setup Wizard

Custom Configuration
When this wizard closes, you can configure the selected services in the Routing and Remote Access console.

Select the services that you want to enable on this server.

- ☒ VPN access
- ☐ Dial-up access
- ☐ Demand-dial connections (used for branch office routing)
- ☐ NAT
- ☒ LAN routing

Routing and Remote Access Server Setup Wizard

Completing the Routing and Remote Access Server Setup Wizard

You have successfully completed the Routing and Remote Access Server Setup wizard.

Summary of selections:

VPN access

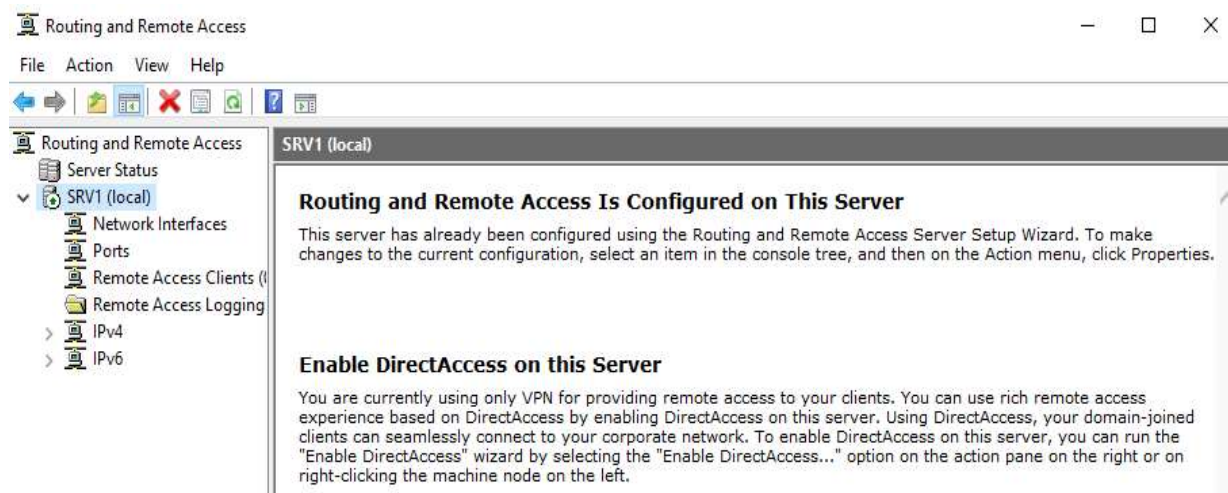
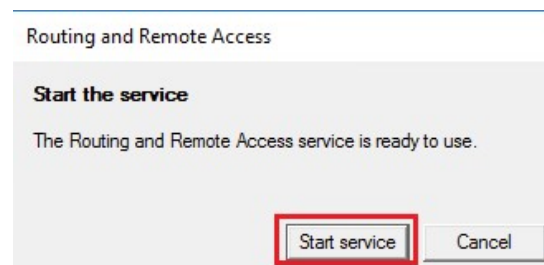
After you close this wizard, configure the selected services in the Routing and Remote Access console.

To close this wizard, click Finish.

< Back Next > < Back Finish Cancel

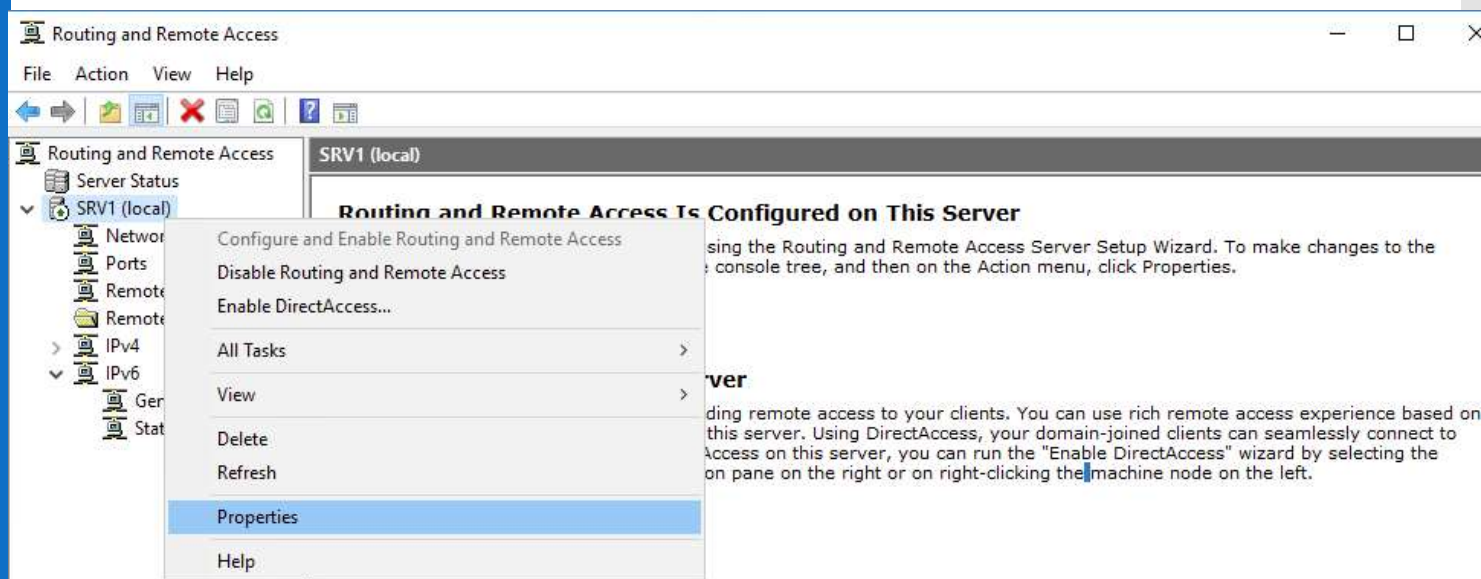
Krok 8

::Konfigurácia role VPN



Krok 8

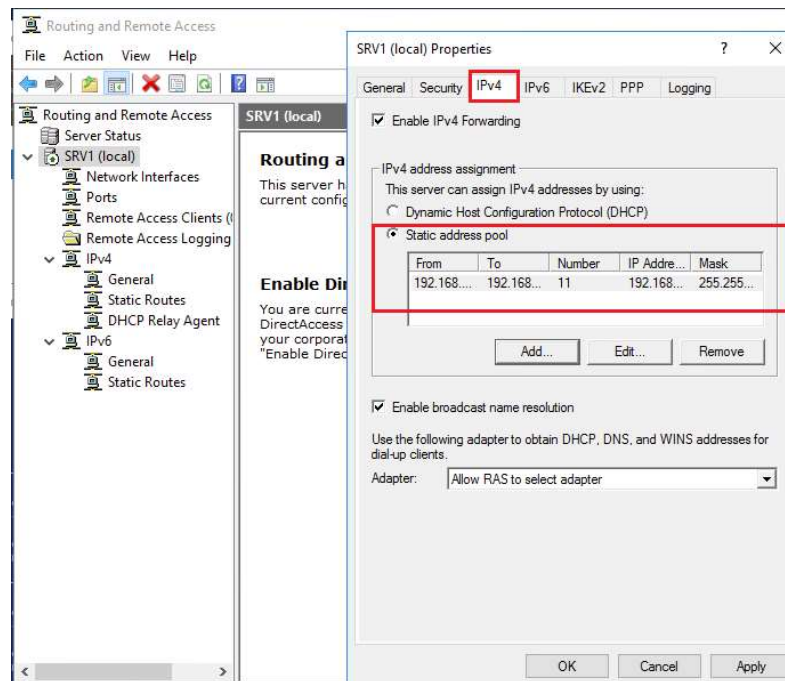
::Konfigurácia role VPN



Krok 8.5

pokiaľ nechcete použiť DHCP rolu na DC!

::Konfigurácia role VPN: IP pool verA (musí byť s rovnakej siete ako "čierna")



Po dokončení chod'rovno na krok 9 !

Krok 8.6

pokiaľ chcete použiť DHCP rolut na DC!

::Konfigurácia role VPN: IP pool verB (len ak si predchádzajúci krok 8.5 nevykonal !)

The screenshot shows the Windows Routing and Remote Access console. In the left-hand tree view, the path is: Routing and Remote Access > SRV1 (local) > IPv4 > DHCP Relay Agent. This path is highlighted with a red box. The main pane displays the 'DHCP Relay Agent' status table, which is also highlighted with a red box. The table has four columns: Interface, Relay mode, Requests received, and Replies received. The data row shows 'NIC-EXTERNAL', 'Enabled', '9', and '0' respectively. Below the table, the 'DHCP Relay Agent Properties' dialog box is open to the 'General' tab. In this dialog, the 'Server address' field contains the text '<P.DHCP SERV>' and is highlighted with a red box. The 'Add' button next to the field is also visible.

Interface	Relay mode	Requests received	Replies received
NIC-EXTERNAL	Enabled	9	0

DHCP Relay Agent Properties - General tab

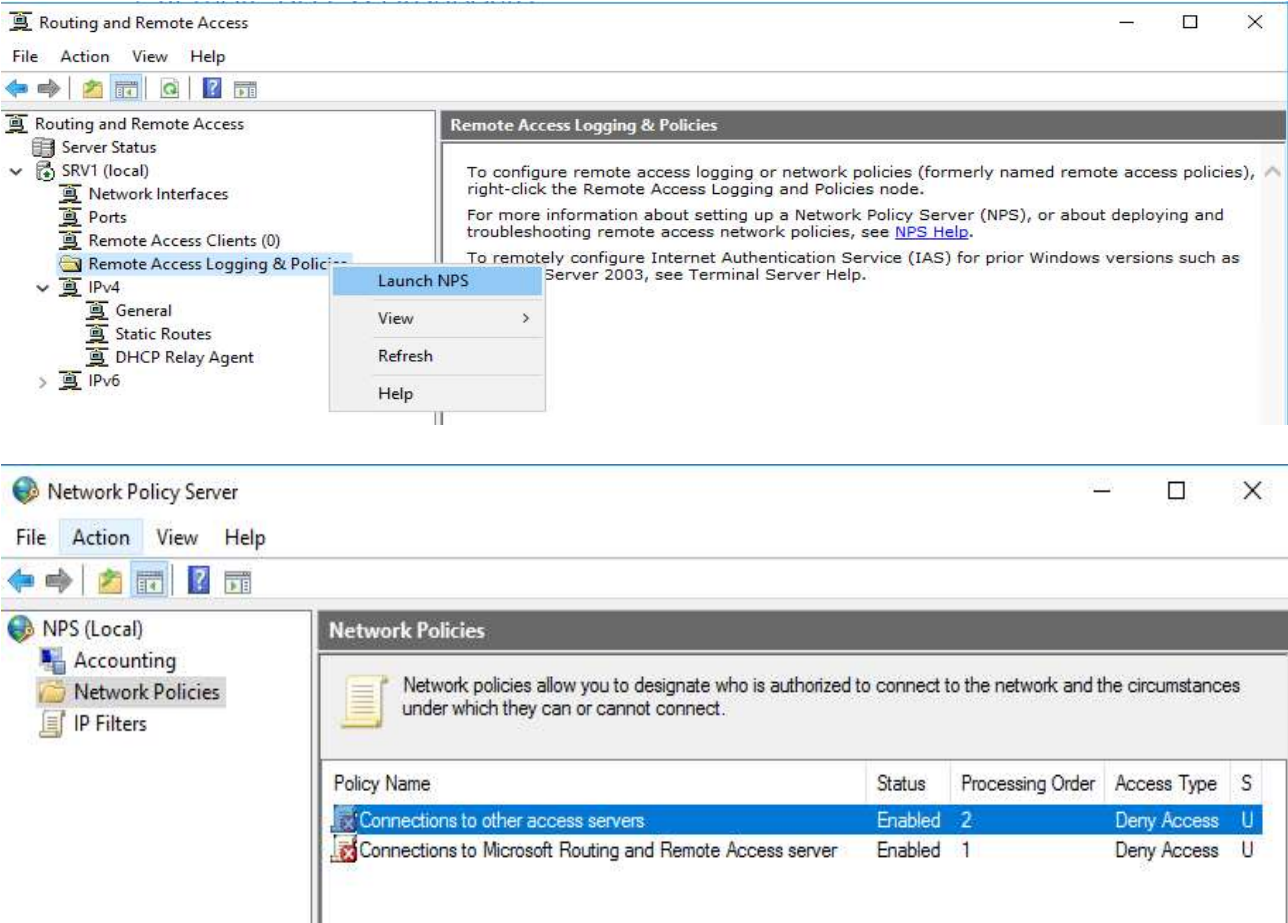
Dynamic Host Configuration Protocol (DHCP) Global

The DHCP relay agent sends messages to the server addresses listed below.

Server address: <P.DHCP SERV> [Add] [Remove]

Krok 9

::Konfigurácia Access Policy cez NPS



Krok 9

::Konfigurácia Access Policy cez NPS

Network Policy Server

File Action View Help

NPS (Local)

Accounting

Network Policies

IP Filters

New

Export List

View

Refresh

Help

Network Policies

Network policies allow you to designate who is authorized to connect to the network and the circumstances under which they can or cannot connect.

	Status	Processing Order	Access Type	S
Connections to other access servers	Enabled	2	Deny Access	U
Connections to Microsoft Routing and Remote Access server	Enabled	1	Deny Access	U

New Network Policy

Specify Network Policy Name and Connection Type
You can specify a name for your network policy and the type of connections to which the policy is applied.

Policy name:

UserVPNAllowAccess

Network connection method

Select the type of network access server that sends the connection request to NPS. You can select either the network access server type or Vendor specific, but neither is required. If your network access server is an 802.1X authenticating switch or wireless access point, select Unspecified.

☒ Type of network access server:

Remote Access Server(VPN-Dial up)

☐ Vendor specific:

10

Krok 9

::Konfigurácia Access Policy cez NPS

New Network Policy

 **Specify Conditions**

Specify the conditions that determine whether this network policy is evaluated for a connection request. A minimum of one condition is required.

Conditions:

Condition	Value
-----------	-------

Condition description:

Add... Edit... Remove

Previous Next Finish Cancel

Krok 9

::Konfigurácia Access Policy cez NPS

New Network Policy

Specify Conditions

Specify the conditions that determine whether this network policy is evaluated for a connection request. A minimum of one condition is required.

Conditions:

Condition	Select condition
	Select a condition, and then click Add.
	Groups
	Windows Groups The Windows Groups condition specifies that the connecting user or computer must belong to one of the selected groups.
	Machine Groups The Machine Groups condition specifies that the connecting computer must belong to one of the selected groups.
	User Groups The User Groups condition specifies that the connecting user must belong to one of the selected groups.
	Day and time restrictions
	Day and Time Restrictions Day and Time Restrictions specify the days and times when connection attempts are and are not allowed. These restrictions are based on the time zone where the NPS server is located.
	Connection Properties

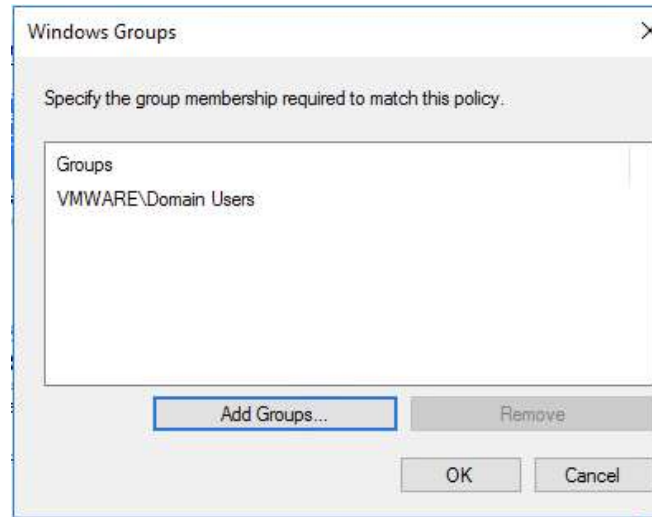
Condition description

Add... Cancel

Add... Edit... Remove

Krok 9

::Konfigurácia Access Policy cez NPS



New Network Policy



Specify Conditions

Specify the conditions that determine whether this network policy is evaluated for a connection request. A minimum of one condition is required.

Conditions:

Condition	Value
 Windows Groups	VMWARE\Domain Users

Condition description:

The Windows Groups condition specifies that the connecting user or computer must belong to one of the selected groups.

Krok 9

::Konfigurácia Access Policy cez NPS

New Network Policy



Specify Access Permission

Configure whether you want to grant network access or deny network access if the connection request matches this policy.

☒ Access granted

Grant access if client connection attempts match the conditions of this policy.

☐ Access denied

Deny access if client connection attempts match the conditions of this policy.

☐ Access is determined by User Dial-in properties (which override NPS policy)

Grant or deny access according to user dial-in properties if client connection attempts match the conditions of this policy.

Previous

Next

Finish

Cancel

Krok 9

::Konfigurácia Access Policy cez NPS

New Network Policy



Configure Authentication Methods

Configure one or more authentication methods required for the connection request to match this policy. For EAP authentication, you must configure an EAP type.

EAP types are negotiated between NPS and the client in the order in which they are listed.

EAP Types:

Microsoft: Secured password (EAP-MSCHAP v2)

Move Up

Move Down

Add...

Edit...

Remove

Less secure authentication methods:

☒ Microsoft Encrypted Authentication version 2 (MS-CHAP-v2)

☒ User can change password after it has expired

☐ Microsoft Encrypted Authentication (MS-CHAP)

☐ User can change password after it has expired

☐ Encrypted authentication (CHAP)

☐ Unencrypted authentication (PAP, SPAP)

☐ Allow clients to connect without negotiating an authentication method.

Previous

Next

Finish

Cancel

Krok 9

::Konfigurácia Access Policy cez NPS Všetky ďalšie nastavenie nechajte default

New Network Policy

**Configure Constraints**

Constraints are additional parameters of the network policy. If a constraint is not matched by the connection request, network access is denied. If you do not want to configure constraints, click Cancel.

New Network Policy

**Configure Settings**

NPS applies settings to the connection request if all of the network policy's constraints are matched.

New Network Policy

**Completing New Network Policy**

You have successfully created the following network policy:
UserVPNAllowAccess

Configure the constraints for this network policy.
If all constraints are not matched by the connection request, network access is denied.

Constraints:

Constraints

Idle Timeout

Session Timeout

Called Station ID

Day and time restrictions

NAS Port Type

Specify the maximum time in minutes that a session can last before the user is disconnected.

☐ Disconnect after the maximum session time.

Configure the settings for this network policy.
If conditions and constraints match the connection request and the policy grants access.

Settings:

RADIUS Attributes

Standard

☒ Vendor Specific

To send additional attributes to RADIUS clients then click Edit. If you do not configure an attribute, your RADIUS client documentation for required attributes.

Name	Value
Framed-Protocol	PPP
Service-Type	Framed

Policy conditions:

Condition	Value
Windows Groups	TESTLAB\Domain Users OR TESTLAB\Domain Admins

Policy settings:

Condition	Value
Authentication Method	EAP OR MS-CHAP v2 OR MS-CHAP v2 (User can change password)
Access Permission	Grant Access
Framed-Protocol	PPP
Service-Type	Framed
Ignore User Dial-In Properties	False
Extensible Authentication Protocol Method	Microsoft® Secured password (EAP-MSCHAP v2)

To close this wizard, click Finish.

Previous

Next

Krok 9

::Konfigurácia Access Policy cez NPS

Všetky ďalšie nastavenie nechajte default

New Network Policy

Completing New Network Policy

You have successfully created the following network policy:
USERVPNAllowAccess

Policy conditions:

Condition	Value
Windows Groups	VMWARE\Domain Users

Policy settings:

Condition	Value
Authentication Method	EAP OR MS-CHAP v2 OR MS-CHAP v2 (User can change password after it has expir...
Access Permission	Grant Access
Framed-Protocol	PPP
Service-Type	Framed
Ignore User Dial-In Properties	False
Extensible Authentication Protocol Method	Microsoft: Secured password (EAP-MSCHAP v2)

To close this wizard, click Finish.

Previous Next **Finish** Cancel

Krok 9

::Konfigurácia Access Policy cez NPS

Network Policy Server

File Action View Help

NPS (Local)

- Accounting
- Network Policies
- IP Filters

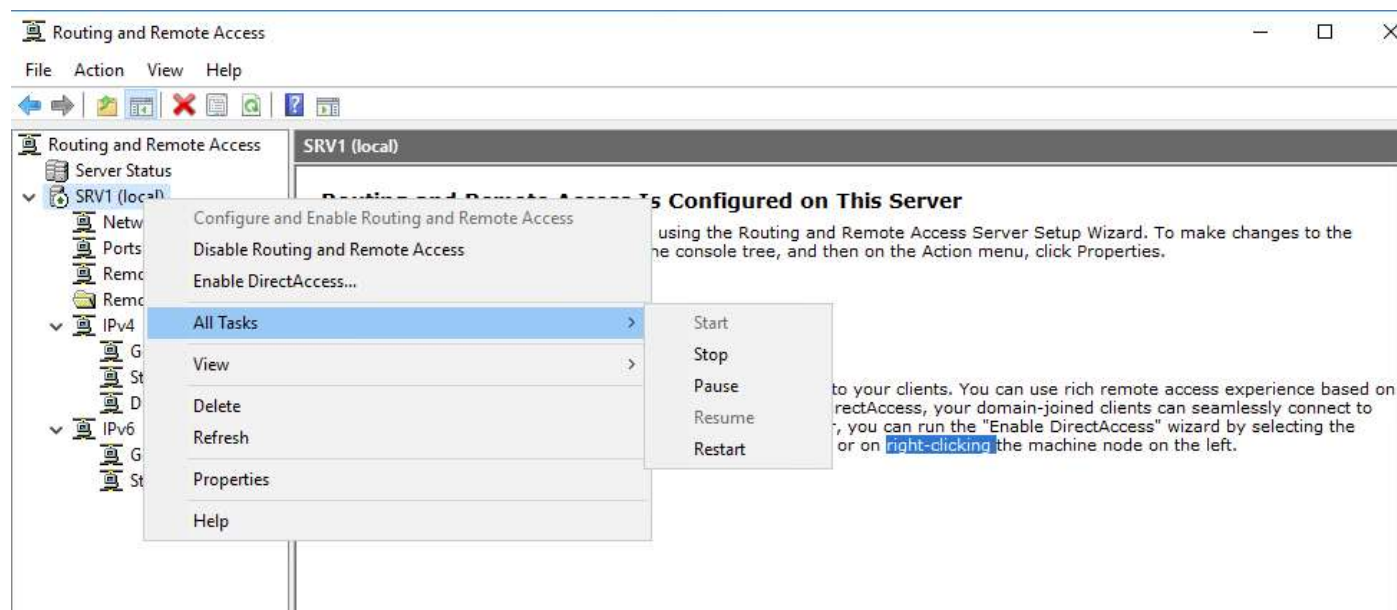
Network Policies

Network policies allow you to designate who is authorized to connect to the network and the circumstances under which they can or cannot connect.

Policy Name	Status	Processing Order	Access Type	S
UserVPNAllowAccess	Enabled	3	Grant Access	R
Connections to other access servers	Enabled	2	Deny Access	U
Connections to Microsoft Routing and Remote Access server	Enabled	1	Deny Access	U

Krok 9

::Restart aplikácie

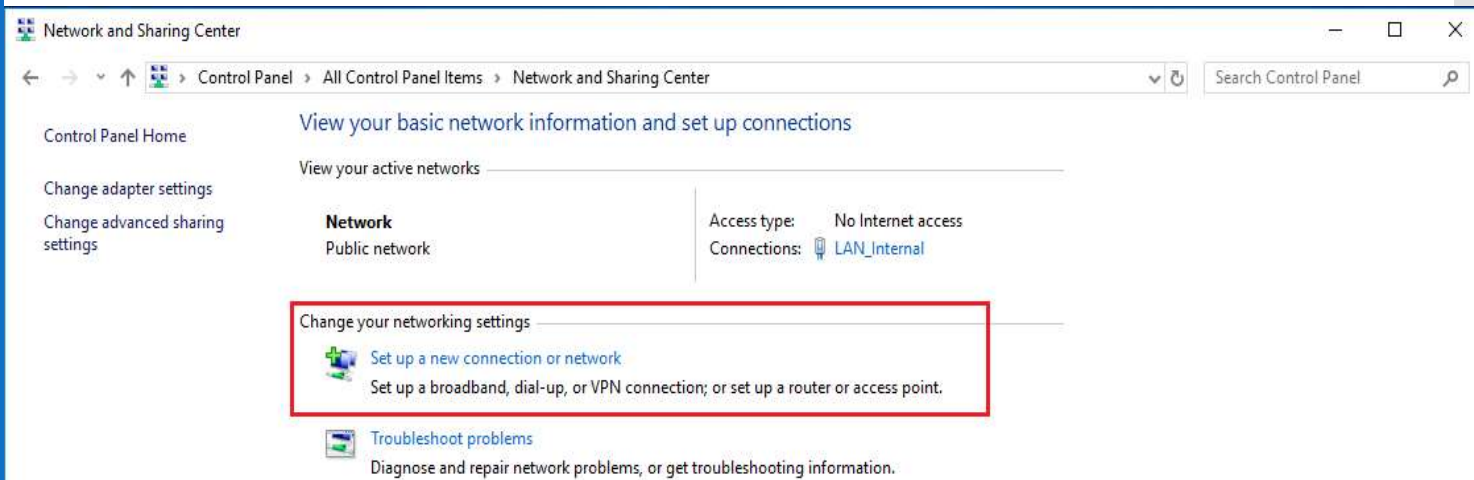


Client cfg

::Pokračujte na konfiguráciu klienta

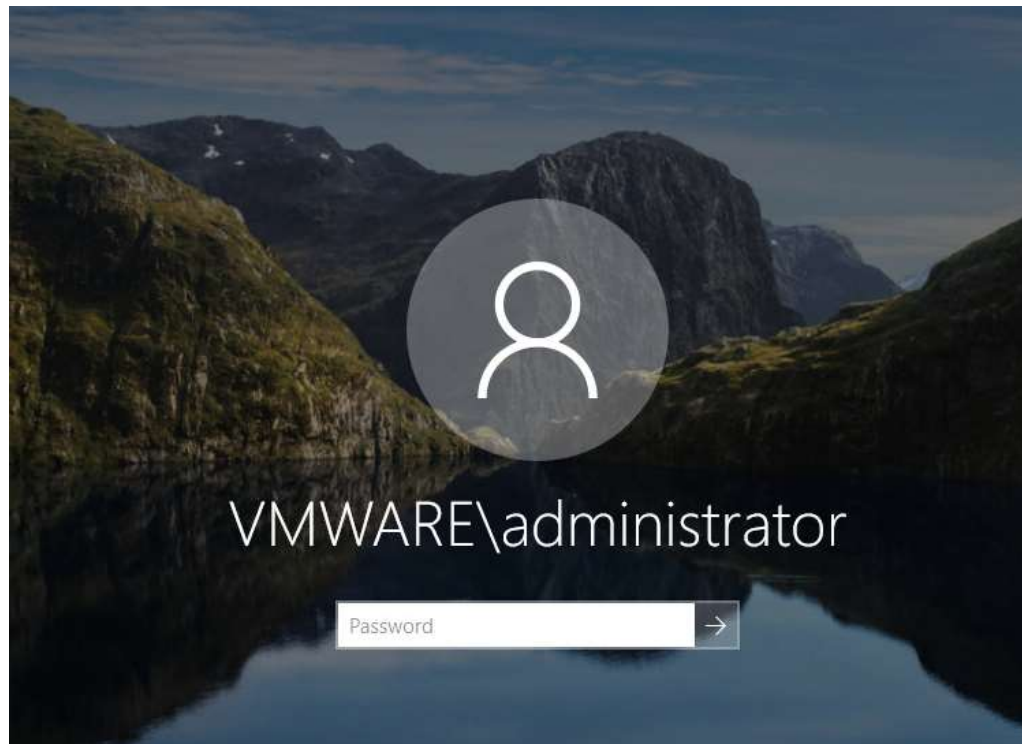
Krok 10

::Klient nastavenie (CLI vm)



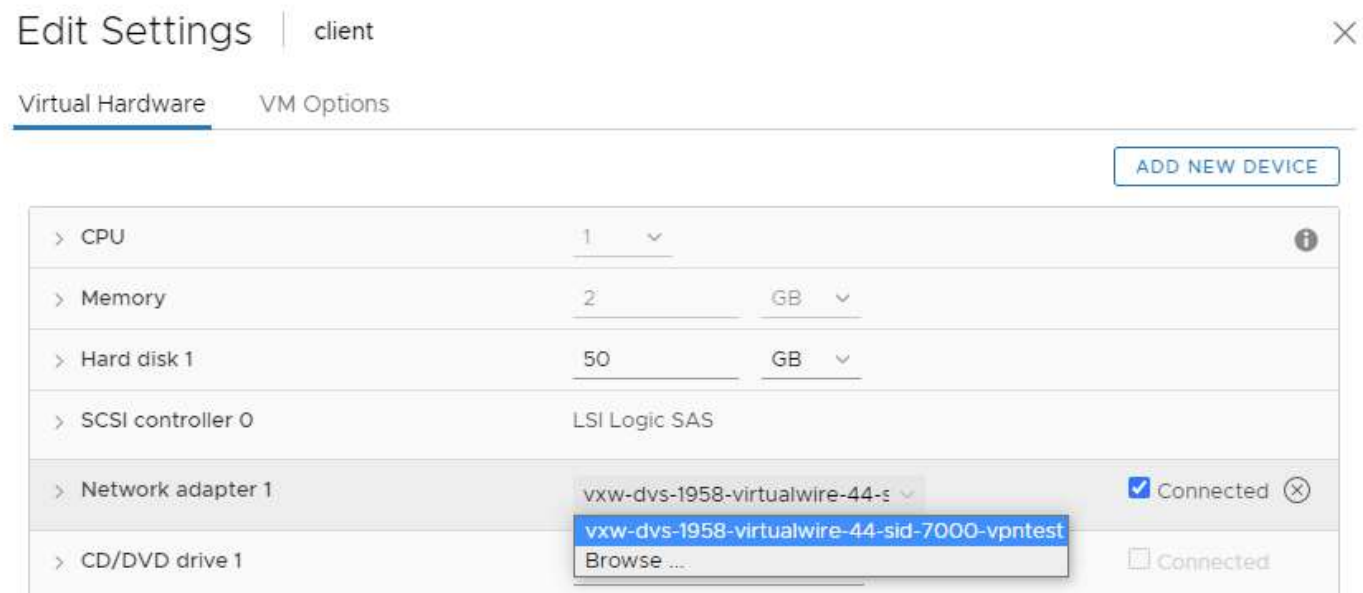
Krok 10

::Login s domain accountom <domain>\Administrator



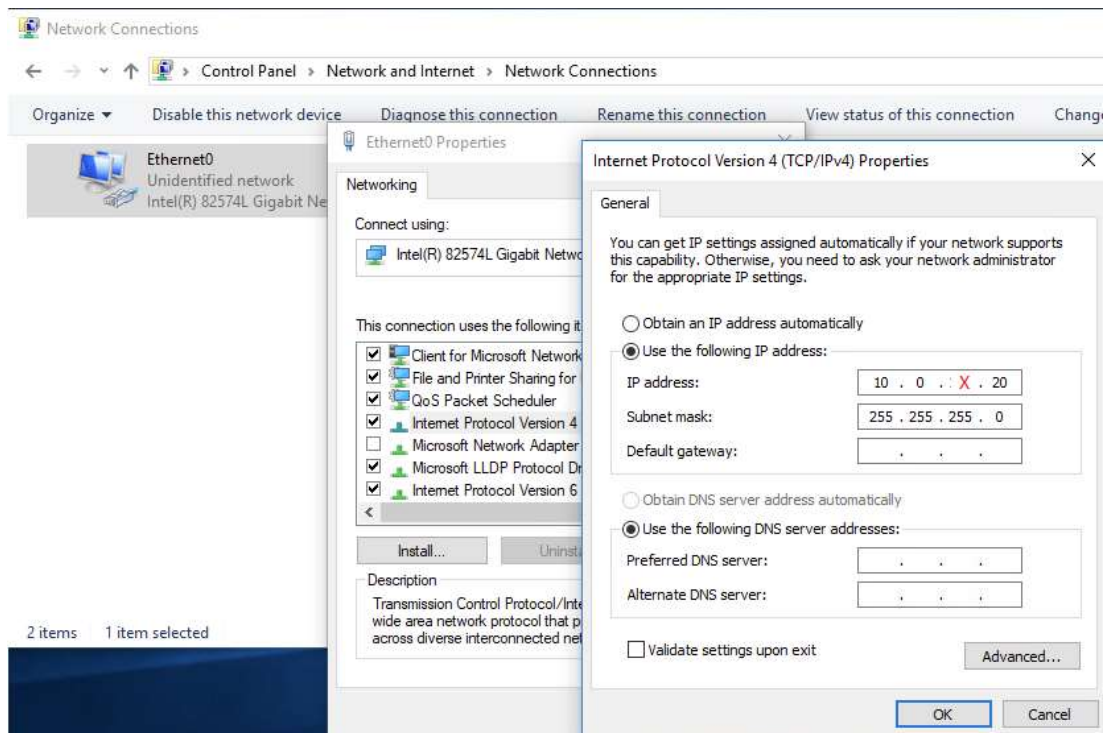
Krok 10

::Po úspešnom login, zmeniť network na public (oranžová) cez vmware konzolu nad client VM



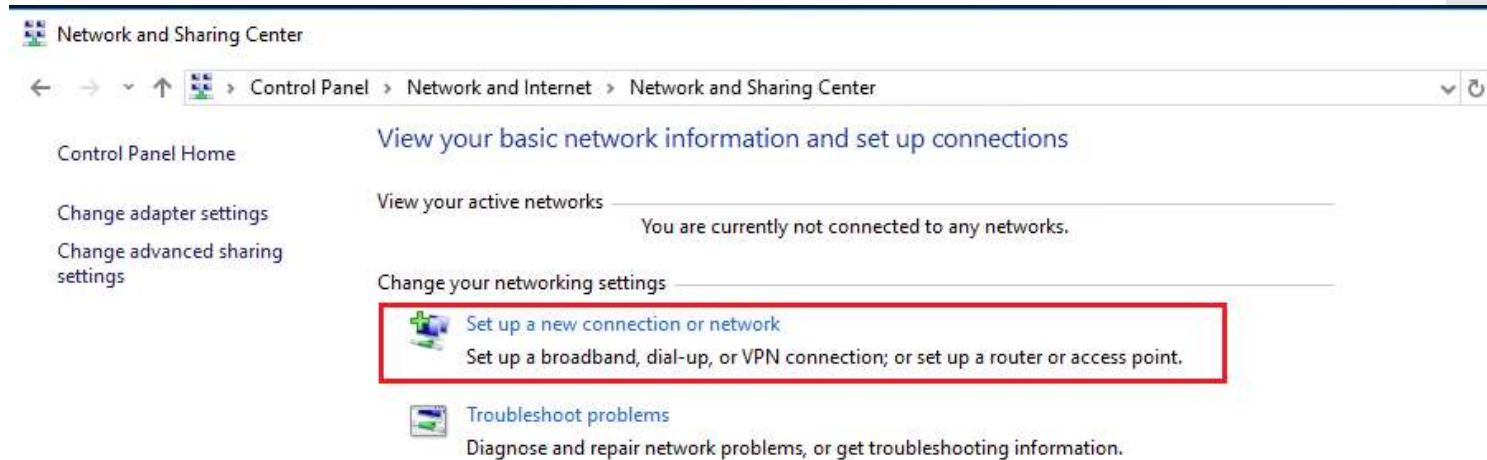
Krok 10

::Change IP klienta na IP z verejnej siete (oranžová), aby vedeli komunikovať client a VPN server ext NIC



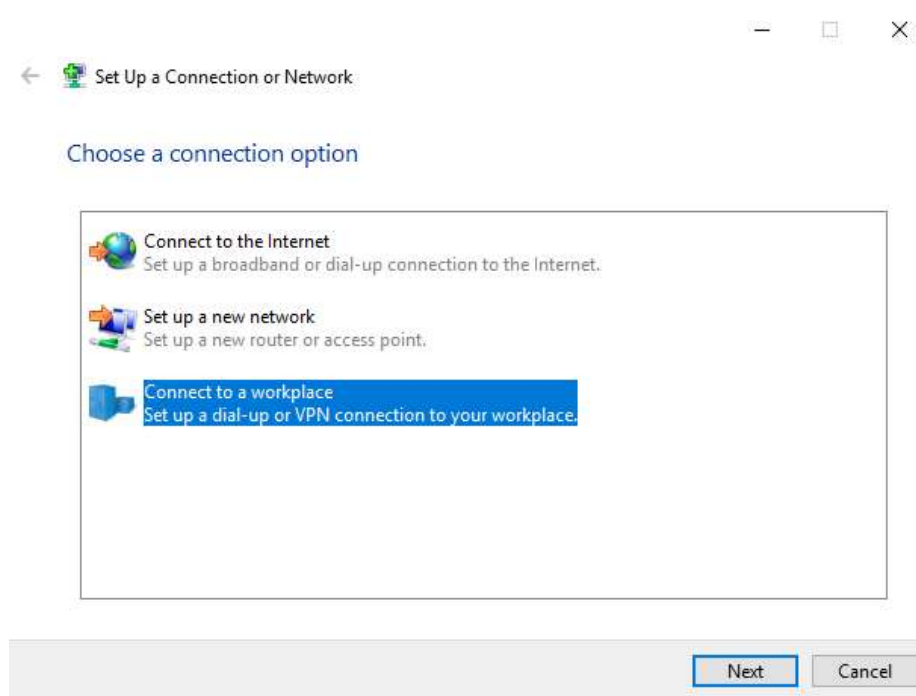
Krok 10

::Klient nastavenie (CLI vm)



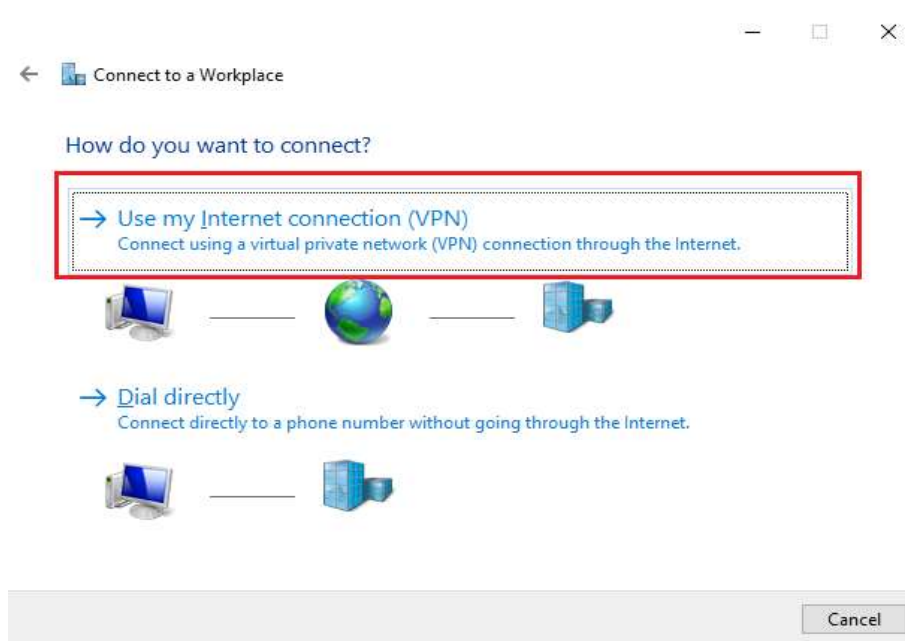
Krok 10

::Klient nastavenie (CLI vm)



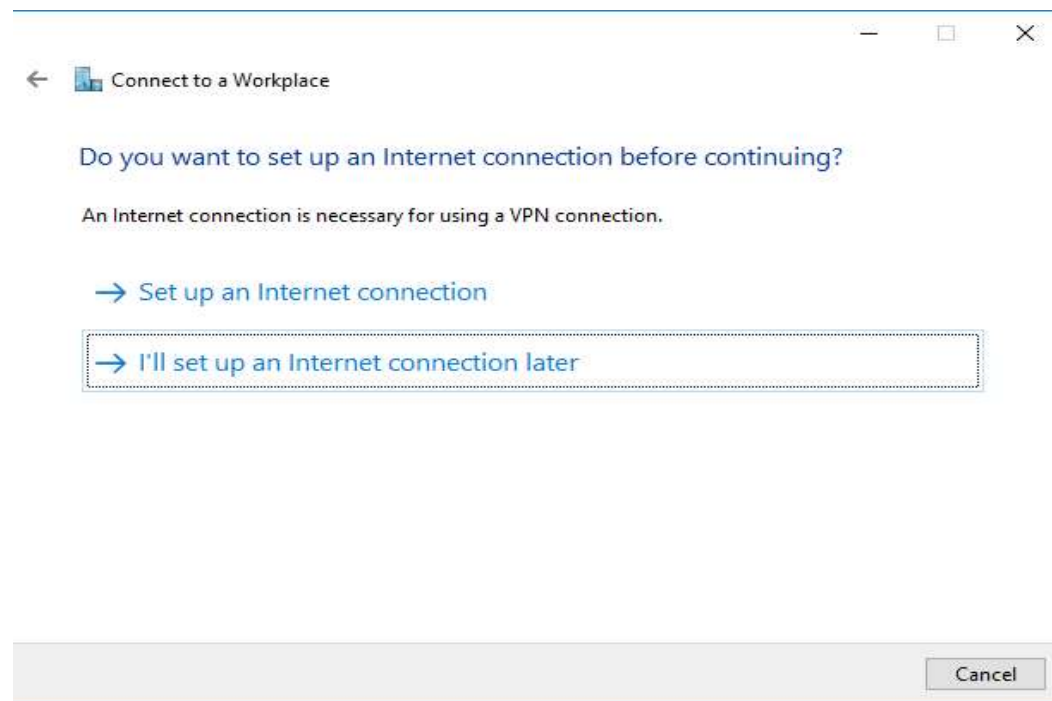
Krok 10

::Klient nastavenie (CLI vm)



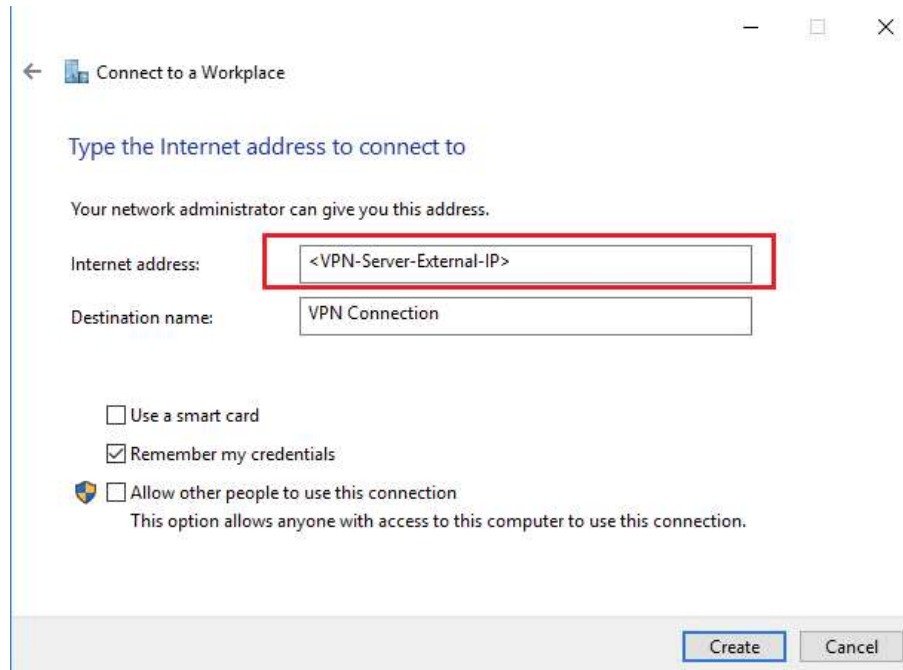
Krok 10

::Klient nastavenie (CLI vm)



Krok 10

::Klient nastavenie (CLI vm)



←  Connect to a Workplace

Type the Internet address to connect to

Your network administrator can give you this address.

Internet address:

Destination name:

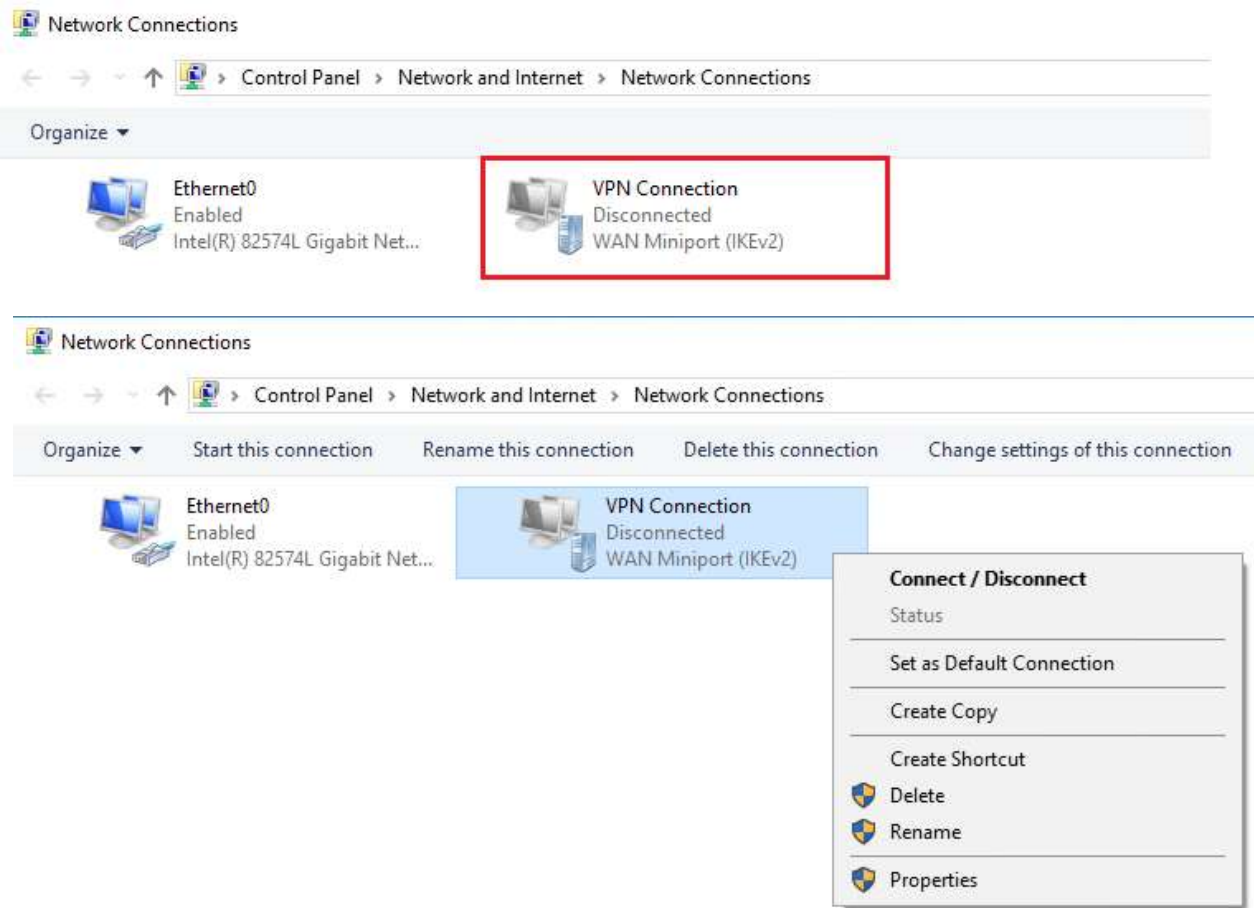
☐ Use a smart card

☒ Remember my credentials

 ☐ Allow other people to use this connection
This option allows anyone with access to this computer to use this connection.

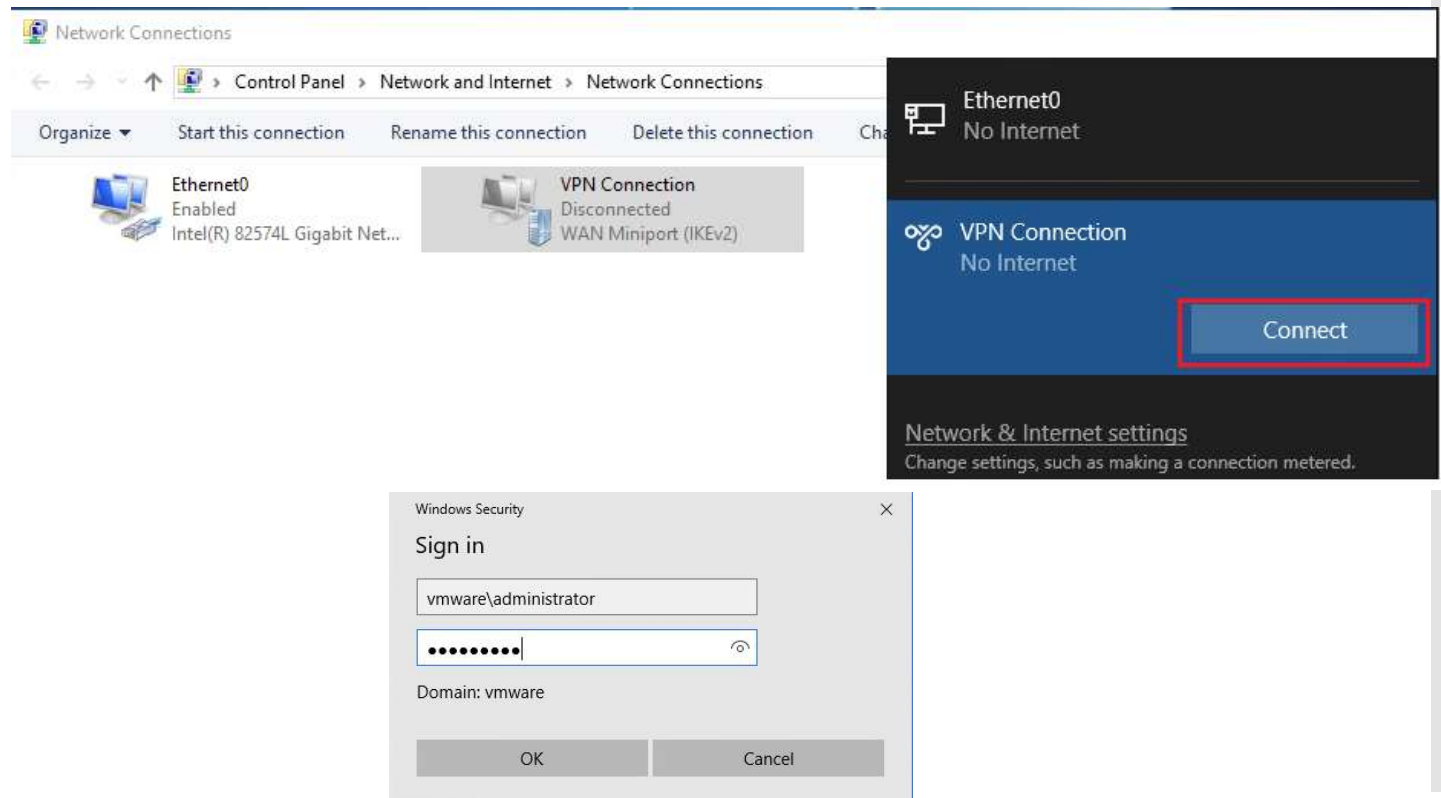
Krok 10

::Pokus o pripojenie do VPN cez nový virtuálny VPN adaptér



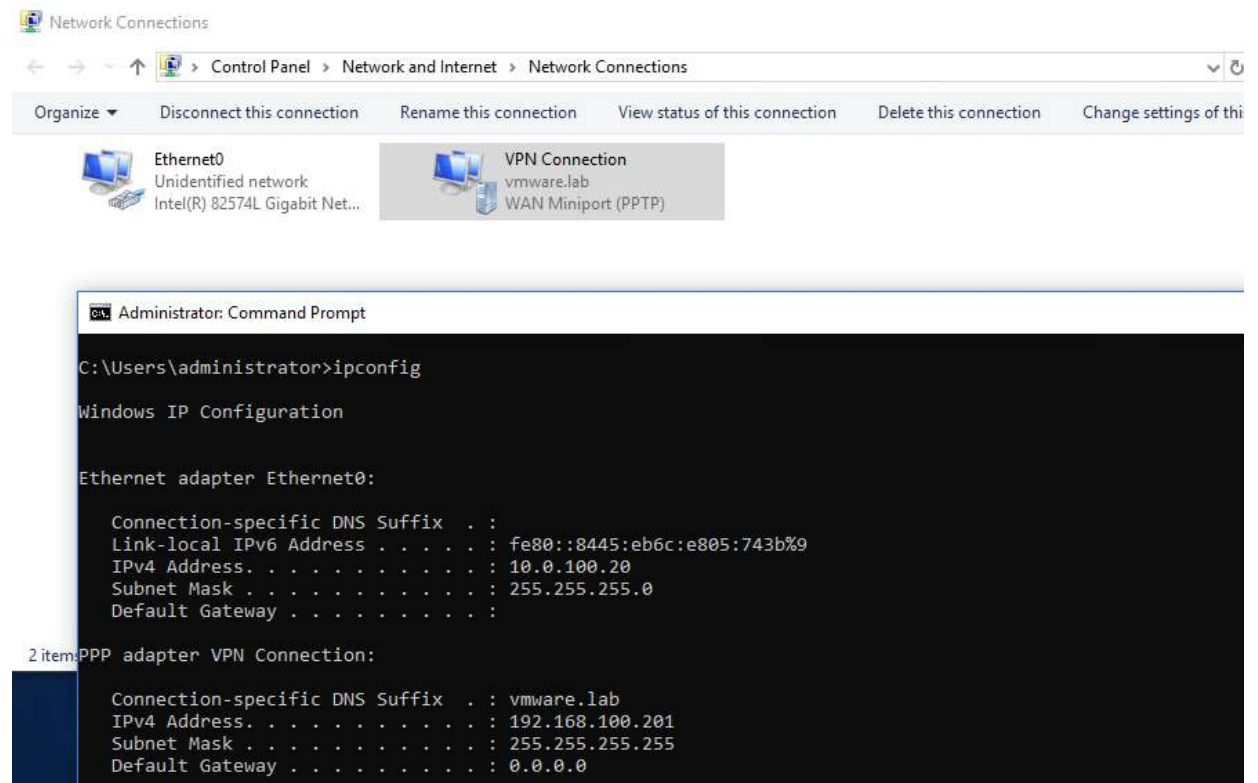
TEST VPN pripojenia

::Pripojiť sa z CLI do korporátnej siete cez externý adaptér



TEST VPN pripojenia

::Po úspešnom pripojení ste v sieti



The image shows two screenshots from a Windows operating system. The top screenshot is the 'Network Connections' window in the Control Panel. It displays two network adapters: 'Ethernet0' (Unidentified network, Intel(R) 82574L Gigabit Net...) and 'VPN Connection' (vmware.lab, WAN Miniport (PPTP)). The 'VPN Connection' is highlighted. The bottom screenshot is an 'Administrator: Command Prompt' window showing the output of the 'ipconfig' command. It displays the IP configuration for both the Ethernet adapter and the VPN Connection. The Ethernet adapter has a Link-local IPv6 Address of fe80::8445:eb6c:e805:743b%9, an IPv4 Address of 10.0.100.20, a Subnet Mask of 255.255.255.0, and a Default Gateway. The VPN Connection has a Connection-specific DNS Suffix of vmware.lab, an IPv4 Address of 192.168.100.201, a Subnet Mask of 255.255.255.255, and a Default Gateway of 0.0.0.0.

```
Administrator: Command Prompt

C:\Users\administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::8445:eb6c:e805:743b%9
    IPv4 Address. . . . . : 10.0.100.20
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

2 item(s)
PPP adapter VPN Connection:

    Connection-specific DNS Suffix  . : vmware.lab
    IPv4 Address. . . . . : 192.168.100.201
    Subnet Mask . . . . . : 255.255.255.255
    Default Gateway . . . . . : 0.0.0.0
```

TEST VPN pripojenia

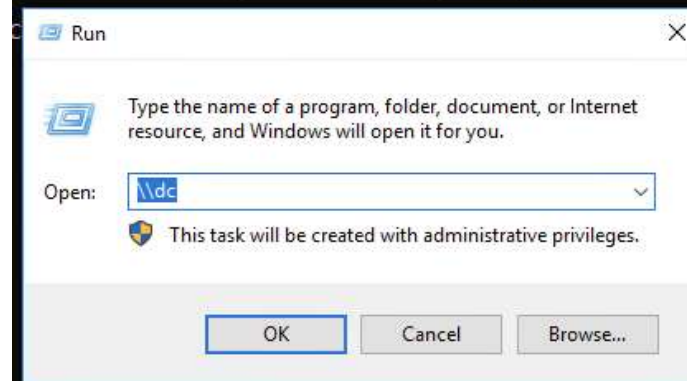
::Po úspešnom pripojení ste v sieti: TEST z klienta

```
Administrator: Command Prompt

C:\Users\administrator>
C:\Users\administrator>ping dc

Pinging dc.vmware.lab [192.168.100.10] with 32 bytes of data:
Reply from 192.168.100.10: bytes=32 time=1ms TTL=127
Reply from 192.168.100.10: bytes=32 time=1ms TTL=127
Reply from 192.168.100.10: bytes=32 time=1ms TTL=127
Reply from 192.168.100.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```



Koniec